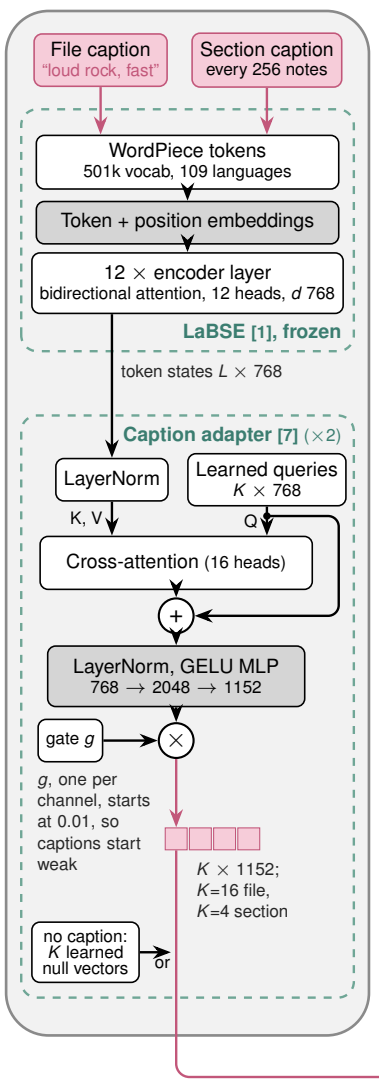
# Captioned music model (427M)

427M trainable parameters: trunk 396M, cascade heads 16M, two caption adapters 13M, embeddings 2M. The LaBSE text encoder (471M) is frozen and not counted.

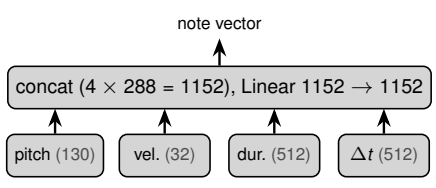
**Status: the music model without caption adapters (464M, full attention) is trained. The caption stage of this layout (F1, with GQA) is planned; caption results here come from 107M pilots. F1 will also add a length-control input (roadmap 11), not drawn here yet; infilling (roadmap 12) comes in a later fine-tune.**

How to read: numbers give the reading order; pink = captions; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part.

## 1 Caption path




## 2 Input embedding

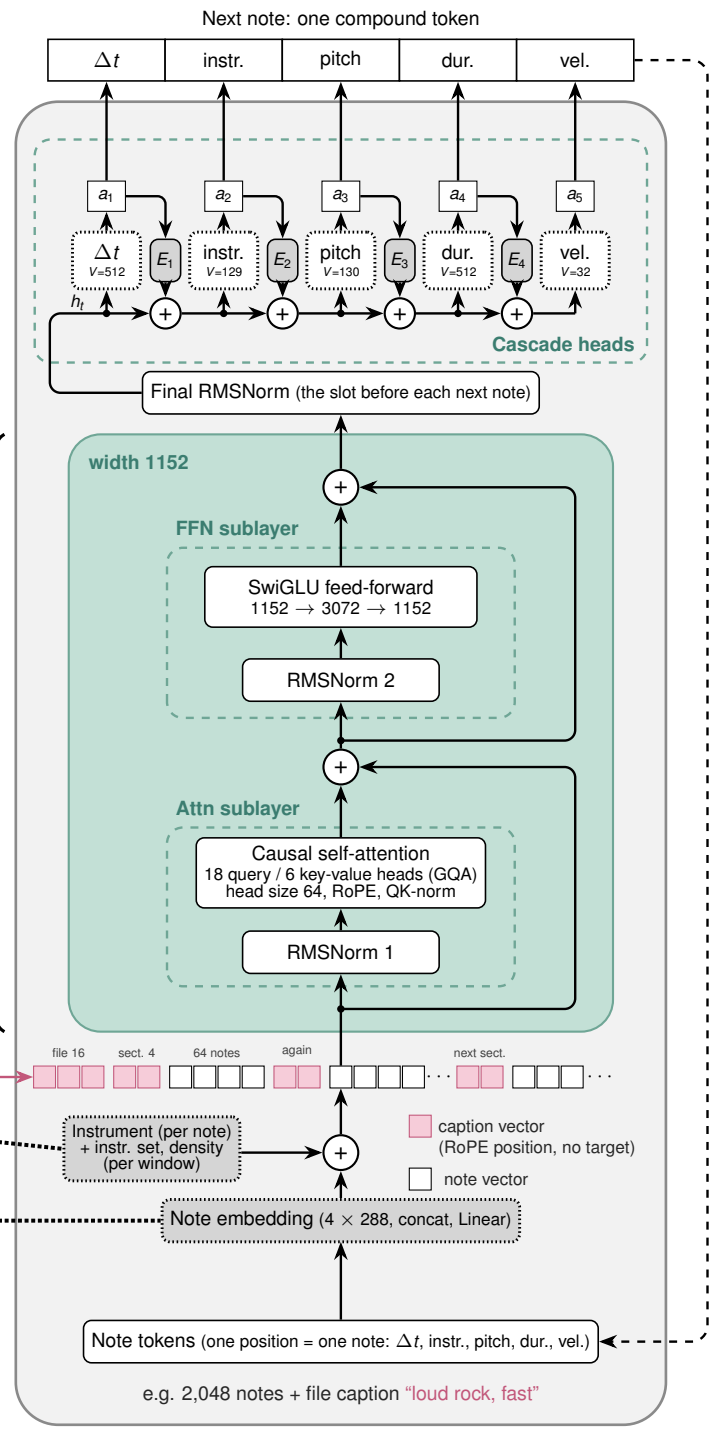



**+ instrument of this note:** a table of 129 vectors (128 General MIDI programs + drums), added to the note vector. The output head *instr.* predicts the next one.

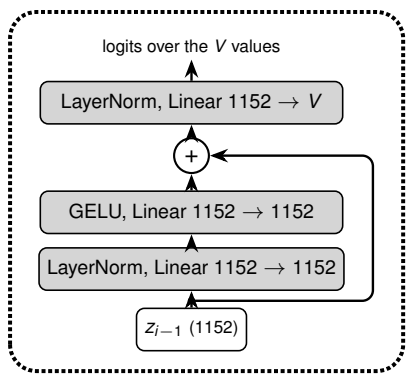
**+ window conditions,** the same vector at every note position: the **instrument set** of the window (129-dim multi-hot → Linear) and its **density** (notes per second in 15 levels + "none" → a table of 16 vectors). Both start at zero and are dropped 15% of the time in training, so "none" stays usable. In generation they come from the user or are measured on the prompt.

**Time is absolute:** $\Delta t$ and durations in 20 ms steps, clipped at 10.22 s. There are no bars, beats or tempo tokens; the metre has to be read from the timing. 2,048 notes are about 1–4 minutes of music.

## 3 Music model




## 4 One cascade head (the 5 dotted boxes, own weights)



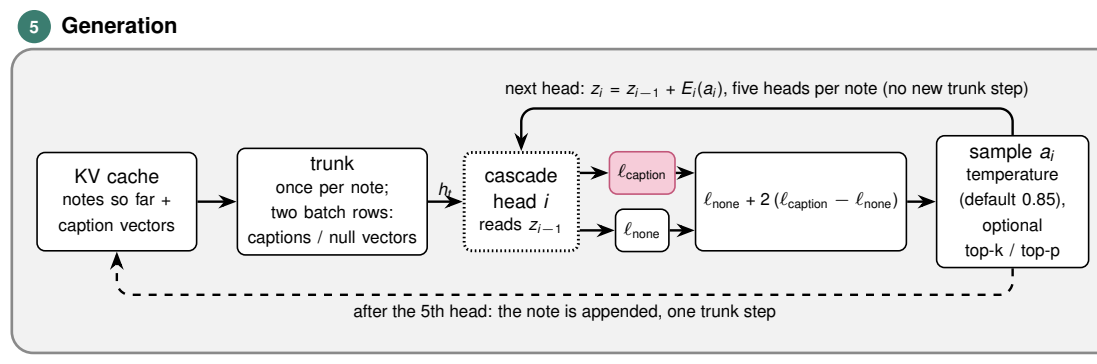

**Why a cascade.** One position predicts all five values of the next note. Head $i$ reads the running state

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

the trunk state $h_t$ plus the embeddings $E$ of the values already chosen. So the note's probability follows the chain rule, $p(\Delta t)\, p(\text{instr.} \mid \Delta t)\, p(\text{pitch} \mid \Delta t, \text{instr.}) \cdots$ (compound tokens as in [6]).

$a_i$ is the true value in training (teacher forcing) and the sampled value in generation. Loss: the sum of the five cross-entropies.

## 5 Generation




Classifier-free guidance [9] with scale 2: every step runs the batch twice as wide, once with the captions and once with null vectors.

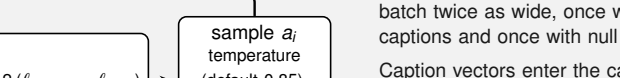
Caption vectors enter the cache as in training: the file caption first, then a section's 4 vectors at its first note and again every 64 notes. Past 2,048 notes the window slides by a quarter.

## 6 Training recipe for section captions

Without these measures the model follows the first section's caption and then ignores later ones.

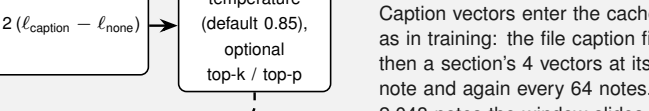
**1. Change-aware captions.** Each section caption names what changed since the section before ("drums enter, louder"). Half of the training windows contain a section change.

**2. Spliced windows (50% of windows).**




File A up to a section boundary, then a section of file B (dashed) with B's caption and no file caption: the caption contradicts the context and has to win.

**3. Caption repeated every 64 notes,** in training and in generation.




The same 4 section vectors are inserted again, so the caption is never far back in the context.

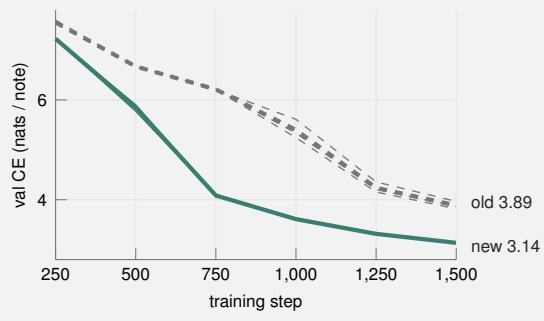
**4. Section words from section statistics.** Loudness, pace and register words use thresholds over sections, not whole files ("loud" = the loudest 20% of sections), and pace is split into rhythm (onsets per second) and texture (notes per onset), so a section caption does not push the music to extremes.

**5. Caption dropout → guidance.** File captions are replaced by null vectors 15% of the time, section captions 20%, so generation can use classifier-free guidance [9] with scale 2.

**Evidence so far** (107M pilot, one seed). Switch test: 6 prompt pairs A/B (e.g. soft / loud), 16 samples each, the section caption changes at note 1,024, guidance 2. The matching attribute (velocity, notes per second, register, drum or string share) is measured per 256 notes; the score is its change after the switch divided by its offset before it (0 = keeps the first style, 1 = a full flip; single prompts reach 1.5). The start model v2_A0: 0.01. 1,500 more steps without measures 2–4 (control): **0.06**, held-out NLL 2.772; splice alone 0.59, repeat alone 0.09. The recipe above (2,700 steps): **0.98**, NLL 2.757. Tested and not used: a counterfactual loss [8] (the true caption must beat the previous one by 0.05 nats on the changed section's first 32 notes): alone 0.13, with splice and repeat 0.85 (0.94 with the section words above), at 1.8× the step time (a second forward). Open cost: holding a caption still overshoots: asked for strings, 34% of samples run away (6.8 notes per onset; 24% and 2.0 with the counterfactual loss). No error bars yet.

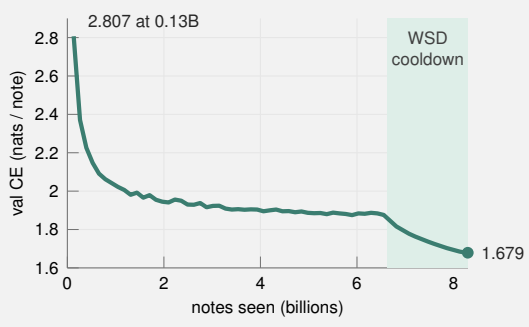
## 7 Why these choices: training curves (teal = what the model uses)

**Trunk block (42M, 98M notes, 3 seeds each)**



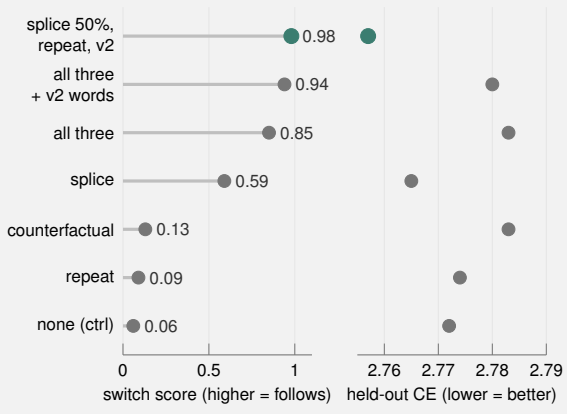

Clean GigaMIDI validation. **New**: RMSNorm, SwiGLU, QK-norm; **old** (dashed): LayerNorm, GELU; both RoPE and Muon. Thin lines = seeds, thick = mean; −0.75 at the end.

**Base model pretraining (464M, 8.3B notes)**



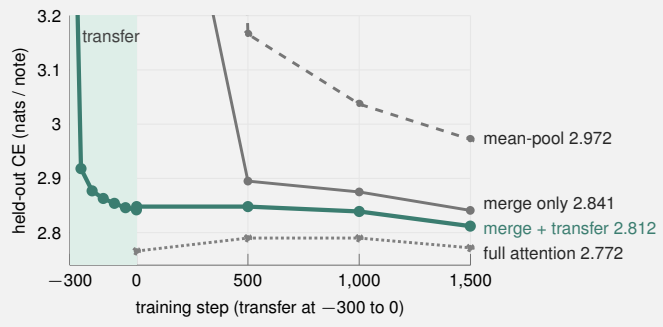

Clean GigaMIDI validation every 2,000 steps (65,536 notes per step); constant LR, then the $1 - \sqrt{\cdot}$ cooldown in the last 20%. One run on one RTX 5090, ≈41 h.

**Section-caption measures (107M pilot, 1,500–2,700 steps)**




Each row continues the same model (v2_A0) for 1,500 steps with one measure or several; the last row runs 2,700 steps, the wall time of "all three" (no second forward). v2 words: measure 4 of section 6. Switch score as in section 6; one seed, no error bars. **Teal** = the recipe in use.

**GQA conversion (107M pilot, 12 → 4 key / value heads)**




v2_A0 converted, then 1,500 steps on the section recipe's data, LR $10^{-4}$, no section measures; the dotted line is the same 1,500 steps without the conversion. Off the scale at the start: mean-pool 10.15, merge 3.86–3.88. One seed.

**Why GQA [21], and how a trained model is converted**

**What it buys.** Each group of 3 query heads shares one key / value head: the K / V projections and the KV cache shrink 3×, so the 450M has 49.5M fewer parameters (464M → 415M without the caption adapters). Generation reads less per note: the 107M decodes 1.03× faster at 1 row and 2.28× at 64 rows (laptop GPU, bf16); on the Raspberry Pi (int4 weights, int8 cache) the estimate from bytes read per note is ≈1.3× at 1 row and ≈2.1× for 4 candidates with guidance.

**Merge.** Mean-pooling a group's K / V weights starts at CE 10.15. Instead, each group's value heads are merged by a PCA of their outputs weighted by the output projection, and the key heads per RoPE rotation pair by a rank-1 complex merge weighted by the query energy, so RoPE stays exact. Measured on 16 batches of the model's own inputs.

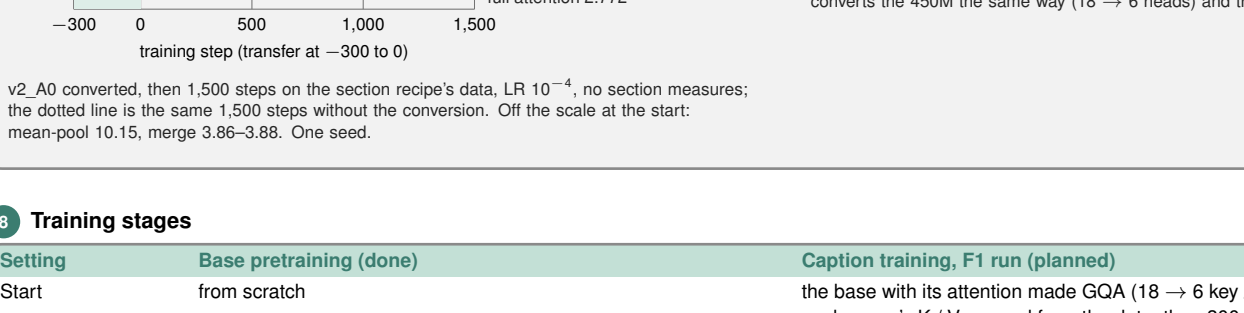
**Transfer.** 300 steps in which only the attention layers train, each to reproduce its own full-attention output (relative MSE) on the original layer inputs, LR $10^{-3}$; the rest of the model is frozen.

**Result.** After 1,500 steps: mean-pool +0.200 vs full attention, merge only +0.069, merge + transfer +0.040 and still closing. F1 converts the 450M the same way (18 → 6 heads) and trains ≈30× longer.

## 8 Training stages

| Setting | Base pretraining (done) | Caption training, F1 run (planned) | Fine-tune on Ukrainian songs |
|---|---|---|---|
| Start | from scratch | the base with its attention made GQA (18 → 6 key / value heads: each group's K / V merged from the data, then 300 steps of per-layer attention transfer) + the two new caption adapters; also planned, not drawn: a length-control input (roadmap 11) | the base (done, no captions); the F1 model with captions (F2, planned) |
| Data | GigaMIDI [12] 0.4, Aria-MIDI [13] 0.3, Discover MIDI [19] 0.3; 8.3B notes (1.8 epochs on average; Aria ≈4.5) | captioned MIDI in 3 tiers: MidiCaps [14] / XMIDI [20] / GigaMIDI metadata, Aria-MIDI metadata, the rest with feature captions; 2.5–4B notes | 5,358 songs transcribed from audio (section 9; 4,811 train, 547 val) 0.75 + GigaMIDI replay 0.25 against forgetting |
| Captions | none: only the instrument-set and density conditions | templates filled from metadata (genre, mood, artist, composer) and from features computed from the notes (tempo, loudness, register, instruments), 30% in Ukrainian; MidiCaps' LLM-written captions; section captions from each section's own notes | none in the done run; F2 adds them |
| Window | 2,048 notes (≈1–4 min) | 2,048 notes + 16 file vectors + 4 per caption arrival (section start and every 64 notes): 2,212 positions (37 caption slots per window, fixed shape; ≈2,192 used) | 2,048 notes |
| Batch | 65,536 notes per step (32 windows) | same | 24,576 notes per step (12 windows) |
| Optimizer, LR | Muon [10] for weight matrices, AdamW for the rest; one LR $10^{-3}$ (Muon updates scaled to AdamW's size) | 8-bit AdamW, LR $10^{-4}$, adapters 10× (as in the 107M pilots) | 8-bit AdamW, LR $2 \cdot 10^{-4}$ |
| Schedule | WSD [11]: warmup, stable (constant LR), $1 - \sqrt{\cdot}$ decay over the last 20% | same; planned: captioned data weighted up in the decay (roadmap 13) | same |
| Steps | 126,582 (≈41 h on one RTX 5090) | ≈38–61k | 1,500 |
| Precision | bf16 with fp8 matmuls, torch.compile | same | same |

## 9 Audio → MIDI: how songs become training notes (fine-tune data)




Every step adds errors: stem bleed, missed or extra notes, octave and timing errors, drum classes mixed up, loudness guessed from activations; stems quieter than −30 dB vs the mix are dropped. The model learns these errors as if they were music.

## 10 Glossary

| | |
|---|---|
| **$\Delta t$** | time since the previous note started (20 ms steps, 512 values) |
| **instr.** | instrument: General MIDI program 0–127 or drums (129 values) |
| **pitch** | MIDI pitch 0–127 + start / end of piece (130 values) |
| **dur.** | note length (20 ms steps, 512 values) |
| **vel.** | velocity: how hard the note is played (32 levels) |
| **compound token** | one sequence position = one whole note |
| **caption vector** | a caption-adapter output placed in the sequence; no target of its own, but the last vector of a section caption predicts the next note |
| **null vectors** | learned stand-ins for "no caption" |
| **teacher forcing** | training feeds the true earlier values, not samples |
| **KV cache** | keys and values of earlier positions kept between steps (GQA: 6 heads per layer, a third of 18) |
| **multi-hot** | a 0/1 vector with a 1 for every instrument present |
| **stem** | one separated part of a recording (vocals, bass, drums, ...) |
| **RoPE [2]** | rotary position embedding: positions as rotations of queries and keys |
| **RMSNorm [3]** | layer normalisation by the root mean square only |
| **SwiGLU [4]** | feed-forward layer with a SiLU-gated linear unit |
| **QK-norm [5]** | queries and keys normalised before the dot product |
| **LaBSE [1]** | multilingual sentence encoder (109 languages) |
| **NLL** | negative log-likelihood of the true notes |
| **nats** | units of NLL with the natural log |
| **BCE** | binary cross-entropy (two-class loss) |
| **CFG [9]** | classifier-free guidance: push logits away from the caption-free prediction |
| **WSD [11]** | warmup, stable, decay: LR schedule with a short final cooldown |
| **Muon [10]** | optimizer that orthogonalises the update of each weight matrix |
| **Demucs [15]** | neural source separation: a song into 6 stems |
| **GQA [21]** | grouped-query attention: each group of 3 query heads shares one key / value head |

## 11 References (links)


[1] LaBSE: Language-agnostic BERT Sentence Embedding (2020), arXiv:2007.01852
[2] RoFormer: rotary position embedding (2021), arXiv:2104.09864
[3] Root Mean Square Layer Normalization (2019), arXiv:1910.07467
[4] GLU Variants Improve Transformer (2020), arXiv:2002.05202
[5] Query-Key Normalization for Transformers (2020), arXiv:2010.04245
[6] Compound Word Transformer (2021), arXiv:2101.02402
[7] Flamingo: learned-query resampler, gated adapters (2022), arXiv:2204.14198
[8] MuseBarControl: counterfactual loss (2024), arXiv:2407.04331
[9] Classifier-Free Diffusion Guidance (2022), arXiv:2207.12598
[10] Muon is Scalable for LLM Training (2025), arXiv:2502.16982
[11] Scaling laws beyond fixed training durations: WSD (2024), arXiv:2405.18392
[12] The GigaMIDI Dataset (2025), arXiv:2502.17726
[13] Aria-MIDI: a dataset of piano MIDI files (2025), arXiv:2504.15071
[14] MidiCaps: MIDI with text captions (2024), arXiv:2406.02255
[15] Demucs: Hybrid Transformers for Music Source Separation (2022), arXiv:2211.08553
[16] basic-pitch: lightweight polyphonic note transcription (2022), arXiv:2203.09893
[17] ADTOF: automatic drum transcription dataset and model (2021), arXiv:2111.11737
[18] EnCodec: High Fidelity Neural Audio Compression (2022), arXiv:2210.13438
[19] Discover MIDI Dataset, Hugging Face dataset
[20] XMusic: the XMIDI dataset (2025), arXiv:2501.08809
[21] GQA: multi-query models from multi-head checkpoints (2023), arXiv:2305.13245